YOUR OPTIMAL RANGES

Testosterone by age: normal isn't the same as optimal

The standard lab range is one wide band for all adult men, so a result can sit comfortably "in range" yet sit low for your age group. Below is the functional target used in *The Testosterone Blueprint* for each decade: a guide, not an official range or a diagnosis.

AGE	STANDARD LAB RANGE	FUNCTIONAL OPTIMAL	WHAT IT MEANS
19–29	264–916	**700–916**	Peak years. A low reading here warrants proper investigation.
30–39	264–916	**600–850**	The gradual ~1%/year decline begins. Lifestyle shows most.
40–49	264–916	**500–750**	Symptoms often appear. Free testosterone matters more.
50–59	264–916	**400–650**	SHBG rises. Always test free T alongside total.
60+	264–916	**350–550**	A "normal" total can still be suboptimal for quality of life.

Values in ng/dL. Multiply by 0.03467 for nmol/L. Lab range = harmonised reference interval (Travison et al., adopted by the Endocrine Society), defined in healthy non-obese men aged 19 to 39; no harmonised range exists for older men, so the same range is shown at every age. Below 264 ng/dL is considered low. Optimal targets from The Testosterone Blueprint, Ch. 2; they are not official reference values.

BEYOND TOTAL TESTOSTERONE

Free T

Range **7–30** ng/dL

The ~2% that's active. Usually calculated from total T, SHBG and albumin. Ranges vary by method and lab, so compare with your own report.

SHBG

Range **16–57** nmol/L

Rises with age (50+: 20–82). High SHBG lowers free T; low SHBG can flag metabolic issues.

Oestradiol

Range **10–40** pg/mL

Men need it too, for bone, libido and mood. Ask for a sensitive/LC-MS assay for accuracy. No age-specific target is established.

Sources: Travison TG et al., *J Clin Endocrinol Metab* 2017 (harmonised reference ranges, adopted by the Endocrine Society); Bhasin S et al., Endocrine Society Clinical Practice Guideline 2018; *The Testosterone Blueprint*, Chapter 2. Educational reference, not a diagnosis. Reviewed November 2026.

BEFORE YOU BOOK

Your blood-test checklist

A “normal” total testosterone alone tells you less the older you get. Take this list to your GP or private lab so you test the right markers, the right way, the first time.

WHAT TO TEST

- [ ] **Total testosterone** — morning sample
- [ ] **SHBG** — binding globulin
- [ ] **Albumin** — needed to calculate free T
- [ ] **Free testosterone** — calculated (Vermeulen)
- [ ] **Oestradiol** — sensitive / LC-MS assay
- [ ] **LH & FSH** — pituitary signals
- [ ] **Prolactin** — if libido is low

WORTH ADDING

- [ ] **TSH** — thyroid, mimics low-T
- [ ] **HbA1c or fasting glucose** — insulin health
- [ ] **Lipid panel** — cardiovascular baseline
- [ ] **Vitamin D** — often low, easily fixed
- [ ] **Ferritin & full blood count** — energy & recovery
- [ ] **Your lab’s own ranges** — note them; they vary

HOW TO TEST RIGHT

- [ ] **Sample between 7–10am.** Testosterone peaks in the morning; an afternoon reading can run 20–25% lower.
- [ ] **Fasted & well-rested.** Avoid testing after heavy alcohol or a poor night’s sleep.
- [ ] **Not during acute illness.** Infection and stress temporarily suppress levels.
- [ ] **Confirm a low result.** Never act on a single reading — repeat before any conclusion.
- [ ] **Insist on a sensitive oestradiol assay.** Standard assays misread male-range levels.
- [ ] **Interpret together.** Read total, free T and SHBG as one picture, alongside symptoms.

PUT IT TO WORK

Plot your results against your age — free & interactive.

Tool → **testosteroneblueprintguide.com/testosterone-levels-by-age** · Full method → **The Testosterone Blueprint**

Disclaimer: This checklist is educational and is not medical advice or a diagnosis. Reference ranges vary by laboratory and assay. Discuss your results and any treatment with a qualified clinician.